Back-off: Linear Interpolation

- Interpolate the trigram, bigram and unigram distributions by a convex combination:

$$\begin{aligned} q(w_i \mid w_{i-2}, w_{i-1}) = \quad & \lambda_1 \times q_{\mathsf{ML}}(w_i \mid w_{i-2}, w_{i-1}) \\ & + \lambda_2 \times q_{\mathsf{ML}}(w_i \mid w_{i-1}) \\ & + \lambda_3 \times q_{\mathsf{ML}}(w_i) \end{aligned}$$

where $\lambda_1 + \lambda_2 + \lambda_3 = 1$, and $\lambda_i \geq 0$ for all i.

- To estimate λ_i, we split the training data into two: compute the n-gram probabilities q_{ML} on one part, and search for the λ_i that yield the highest probability on the other

The Probability of a Corpus

- The log-likelihood of a corpus is the product of the probabilities

$$P(corpus) = \prod_{i=1}^{N} P(s_i)$$

- If the model is a Markov LM (say a tri-gram model), it can be written as: (sub-script is token index, super-script is sentence index)

$$\prod_{i=1}^{N} P(s_i) = \prod_{i=1}^{N} \prod_{j=1}^{length(s_i)} P(w_j^{(i)} | w_{j-1}^{(i)}, w_{j-2}^{(i)})$$

- And if we denote the number of times the trigram (w_1,w_2,w_3) appears in the corpus with c’, we get:

$$\prod_{i=1}^{N} \prod_{j=1}^{length(s_i)} P(w_j^{(i)} | w_{j-1}^{(i)}, w_{j-2}^{(i)}) = \prod_{(w_1, w_2, w_3) \in V^3} P(w_3 | w_1, w_2)^{c'(w_1, w_2, w_3)}$$

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- That is, maximize *L,* the log-likelihood, where $c'(w_1,w_2,w_3)$ is the number of times the trigram (w_1,w_2,w_3) appeared in the validation set

$$L(\lambda_1, \lambda_2, \lambda_3) = log \left(\prod_{(w_1,w_2,w_3) \in V^3} q(w_3|w_1, w_2)^{c'(w_1,w_2,w_3)} \right)$$

- Which yields this (concave) optimization problem:

$$L(\lambda_1, \lambda_2, \lambda_3) = \sum_{w_1,w_2,w_3} c'(w_1, w_2, w_3) \log q(w_3 \mid w_1, w_2)$$

such that $\lambda_1 + \lambda_2 + \lambda_3 = 1$, and $\lambda_i \geq 0$ for all i, and where

$$\begin{aligned} q(w_i \mid w_{i-2}, w_{i-1}) = \quad & \lambda_1 \times q_{\mathsf{ML}}(w_i \mid w_{i-2}, w_{i-1}) \\ & +\lambda_2 \times q_{\mathsf{ML}}(w_i \mid w_{i-1}) \\ & +\lambda_3 \times q_{\mathsf{ML}}(w_i) \end{aligned}$$

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- It is common to take more ls to get a better fit
- Here is one example of how to do that. Define:

$$\Pi(w_{i-2}, w_{i-1}) = \begin{cases} 1 & \text{If Count}(w_{i-1}, w_{i-2}) = 0 \\ 2 & \text{If } 1 \leq \text{Count}(w_{i-1}, w_{i-2}) \leq 2 \\ 3 & \text{If } 3 \leq \text{Count}(w_{i-1}, w_{i-2}) \leq 5 \\ 4 & \text{Otherwise} \end{cases}$$

And maximize:

$$\begin{aligned} q(w_i \mid w_{i-2}, w_{i-1}) = \quad & \lambda_1^{\Pi(w_{i-2},w_{i-1})} \times q_{\text{ML}}(w_i \mid w_{i-2}, w_{i-1}) \\ & + \lambda_2^{\Pi(w_{i-2},w_{i-1})} \times q_{\text{ML}}(w_i \mid w_{i-1}) \\ & + \lambda_3^{\Pi(w_{i-2},w_{i-1})} \times q_{\text{ML}}(w_i) \end{aligned}$$

where $\lambda_1^{\Pi(w_{i-2},w_{i-1})} + \lambda_2^{\Pi(w_{i-2},w_{i-1})} + \lambda_3^{\Pi(w_{i-2},w_{i-1})} = 1$,
and $\lambda_i^{\Pi(w_{i-2},w_{i-1})} \geq 0$ for all i.